Equilibrium

Chemical equilibria are important in numerous biological and environmental processes. For example, equilibria involving O2 molecules and the protein hemoglobin play a crucial role in the transport and delivery of O2 from our lungs to our muscles.

What this unit sets out to do

- identify the dynamic nature of equilibrium involved in physical and chemical processes, state the law of equilibrium, and explain the characteristics of equilibria in both
- write expressions for equilibrium constants, establish a relationship between Kp and Kc, and explain the various factors that affect the equilibrium state of a reaction
- classify substances as acids or bases according to the Arrhenius, Bronsted-Lowry and Lewis concepts, classify acids and bases as weak or strong in terms of their ionization constants, and explain the dependence of degree of ionization on the concentration of the electrolyte and of the common ion
- describe the pH scale for representing hydrogen ion concentration, explain the ionisation of water and its duel role as acid and base, describe the ionic product and pKw for water, appreciate the use of buffer solutions, and calculate the solubility product constant

What equilibrium looks like

- when a liquid evaporates in a closed container, molecules with relatively higher kinetic energy escape the liquid surface into the vapour phase while molecules from the vapour phase strike the liquid surface and are retained, which gives rise to a constant vapour pressure because the number of molecules leaving the liquid equals the number returning
- we say the system has reached the equilibrium state at this stage, however this is not static equilibrium and there is a lot of activity at the boundary between the liquid and the vapour, so at equilibrium the rate of evaporation equals the rate of condensation, and the double half arrows indicate that the processes in both directions are going on simultaneously
- the mixture of reactants and products in the equilibrium state is called an equilibrium mixture, and equilibrium can be established for both physical processes and chemical reactions, the reaction being fast or slow depending on the experimental conditions and the nature of the reactants
- when reactants in a closed vessel at a particular temperature react to give products, the concentrations of the reactants keep decreasing while those of the products keep increasing for some time, after which there is no change in either, and this stage is the dynamic equilibrium, where the rates of the forward and reverse reactions become equal

In this unit

6.1 Equilibrium in Physical Processes

6.2 Equilibrium in Chemical Processes

6.3 Law of Chemical Equilibrium

6.4 Homogeneous Equilibria

6.5 Heterogeneous Equilibria

6.6 Factors Affecting Equilibria

6.7 Ionic Equilibrium

6.8 Acids, Bases and Salts

6.9 Ionization of Acids and Bases

6.10 Solubility Equilibria

Three kinds of reaction, by how far they go

- it is because of the dynamic equilibrium stage that there is no change in the concentrations of the various species in the reaction mixture, and based on the extent to which reactions proceed to reach that state they may be classified in three groups
- those that proceed nearly to completion, where only negligible concentrations of the reactants are left, and in some cases it may not even be possible to detect them experimentally
- those in which only small amounts of products are formed and most of the reactants remain unchanged at the equilibrium stage
- and those in which the concentrations of the reactants and products are comparable when the system is in equilibrium

Why the conditions matter

- the extent of a reaction in equilibrium varies with the experimental conditions such as the concentrations of reactants and the temperature
- optimisation of the operational conditions is very important in industry and laboratory so that equilibrium is favorable in the direction of the desired product
- this unit deals with equilibrium involving physical and chemical processes, along with the equilibrium involving ions in aqueous solutions, which is called ionic equilibrium

6.1 Equilibrium in physical processes, and 6.1.1 solid-liquid equilibrium

Ice and water in a flask

- the characteristics of a system at equilibrium are better understood by examining some physical processes, the most familiar being the phase transformations between solid and liquid, liquid and gas, and solid and gas
- ice and water kept in a perfectly insulated thermos flask at 273 K and atmospheric pressure are in an equilibrium state, and we observe that the masses of ice and water do not change with time and the temperature remains constant
- however the equilibrium is not static, and intense activity can be noticed at the boundary between ice and water, molecules from the liquid colliding against the ice and adhering to it while some molecules of ice escape into the liquid phase, there being no change of mass because the rates of transfer each way are equal

A closed vessel of water with a manometer reading the pressure above it

The pressure climbing as molecules leave the liquid, then levelling off

At that level evaporation and condensation run at the same rate, and the reading is the equilibrium vapour pressure

Figure 6.A Measuring equilibrium vapour pressure of water at a constant temperature

Equilibrium vapour pressure

- the rate of increase in pressure decreases with time due to condensation of vapour into water, and finally leads to an equilibrium condition where there is no net evaporation, the rate of evaporation equalling the rate of condensation
- at equilibrium the pressure exerted by the water molecules at a given temperature remains constant and is called the equilibrium vapour pressure of water, which increases with temperature, and repeating the experiment with methyl alcohol, acetone and ether shows that different liquids have different equilibrium vapour pressures at the same temperature, the liquid with the higher vapour pressure being more volatile and having a lower boiling point
- exposing watch glasses of acetone, ethyl alcohol and water to the atmosphere shows that in every case the liquid eventually disappears, the time taken depending on the nature of the liquid, its amount and the temperature, because when the glass is open the rate of evaporation stays constant but the molecules disperse into the large volume of the room, so the rate of condensation is much less, and it is not possible to reach equilibrium in an open system

Normal boiling point, and 6.1.3 solid-vapour equilibrium

- water and water vapour are in equilibrium at atmospheric pressure and 100 degrees C in a closed vessel, and for any pure liquid at one atmospheric pressure the temperature at which liquid and vapour are at equilibrium is called the normal boiling point, which depends on the atmospheric pressure and so on the altitude of the place, decreasing at high altitude
- placing solid iodine in a closed vessel fills it with violet vapour whose colour deepens with time until the intensity becomes constant, at which stage equilibrium is attained, solid iodine subliming to vapour and the vapour condensing back to solid
- other examples of this kind of equilibrium are camphor and ammonium chloride, each in equilibrium with its own vapour

6.1.4 Equilibrium involving dissolution of solid or gases in liquids

Solids in liquids

- we can dissolve only a limited amount of salt or sugar in a given amount of water at room temperature, and if we make a thick sugar syrup by dissolving sugar at a higher temperature, sugar crystals separate out when the syrup is cooled
- we call it a saturated solution when no more solute can be dissolved at a given temperature, and the concentration of the solute in a saturated solution depends upon the temperature, a dynamic equilibrium existing between the solute molecules in the solid state and in solution, so that the rate of dissolution equals the rate of crystallisation
- the equality of the two rates and the dynamic nature of the equilibrium have been confirmed with radioactive sugar, since dropping some into a saturated solution of non-radioactive sugar leads after some time to radioactivity in both the solution and the solid, the ratio of radioactive to non-radioactive molecules in the solution increasing until it attains a constant value

Gases in liquids

- when a soda water bottle is opened some of the carbon dioxide dissolved in it fizzes out rapidly, which arises from the difference in solubility of carbon dioxide at different pressures, there being an equilibrium between the molecules in the gaseous state and those dissolved in the liquid under pressure
- this equilibrium is governed by Henry's law, which states that the mass of a gas dissolved in a given mass of a solvent at any temperature is proportional to the pressure of the gas above the solvent, and this amount decreases with increase of temperature
- the bottle is sealed under a pressure of gas at which its solubility in water is high, so as soon as it is opened some of the dissolved gas escapes to reach a new equilibrium for the lower partial pressure in the atmosphere, which is how soda water left open turns flat

What all four cases have in common

- for solid and liquid equilibrium there is only one temperature, the melting point, at 1 atm at which the two phases can coexist, and if there is no exchange of heat with the surroundings the mass of the two phases remains constant
- for liquid and vapour equilibrium the vapour pressure is constant at a given temperature, and for dissolution of solids in liquids the solubility is constant at a given temperature
- for dissolution of gases in liquids the concentration of a gas in the liquid is proportional to the pressure of the gas over the liquid

Concentration plotted against time, reactants falling and products rising

The forward rate falling and the reverse rate climbing as they do

Both curves flattening once the two rates are equal

Figure 6.B Attainment of chemical equilibrium.

6.1.5 General characteristics of equilibria involving physical processes

What they have in common

- equilibrium is possible only in a closed system at a given temperature, both the opposing processes occur at the same rate, and there is a dynamic but stable condition
- all measurable properties of the system remain constant, and when equilibrium is attained for a physical process it is characterised by a constant value of one of its parameters at a given temperature
- the magnitude of such a quantity at any stage indicates the extent to which the physical process has proceeded before reaching equilibrium

6.2 Equilibrium in chemical processes, dynamic equilibrium

The same picture for a reaction

- analogous to physical systems, chemical reactions also attain a state of equilibrium, and these reactions can occur both in forward and backward directions, so when the rates of the forward and reverse reactions become equal the concentrations of the reactants and products remain constant, which is the stage of chemical equilibrium
- this equilibrium is dynamic in nature, since it consists of a forward reaction in which the reactants give products and a reverse reaction in which the products give back the original reactants
- taking a general reversible reaction of A and B giving C and D, with the passage of time there is accumulation of the products and depletion of the reactants, which leads to a decrease in the rate of the forward reaction and an increase in the rate of the reverse, until eventually the two occur at the same rate and the system reaches equilibrium
- the reaction can reach that state even if we start with only C and D and no A and B, since equilibrium can be reached from either direction

Haber's experiments

- the dynamic nature of chemical equilibrium can be demonstrated in the synthesis of ammonia by Haber's process, where in a series of experiments he started with known amounts of dinitrogen and dihydrogen at high temperature and pressure and at regular intervals determined the amount of ammonia present, and also the concentrations of the unreacted gases
- after a certain time the composition of the mixture remains the same even though some of the reactants are still present

Two cylinders of water side by side, each with a tube dipped into it

Water carried across a tubeful at a time, in both directions at once

Levels that stop changing while the transfers carry on, which is what dynamic means

Figure 6.C Demonstrating dynamic nature of equilibrium. (a) initial stage (b) final stage after the equilibrium is attained.

A student's activity

- equilibrium, whether in a physical or a chemical process, is dynamic, and this page sets out an activity by which that can be seen rather than taken on trust
- the point of the demonstration is that the quantities stop changing long before the transfers stop, because the two transfers come to run at the same rate

Concentrations plotted against time for the ammonia synthesis

Nitrogen and hydrogen falling, ammonia rising, all three then level

The same flat lines whichever side the mixture started from

Figure 6.D Depiction of equilibrium for the reaction

One run starting from equal amounts of H2 and I2, with HI climbing

Another starting from HI alone, with H2 and I2 climbing instead

Both settling at the same mixture, so equilibrium is reached from either direction

Figure 6.E Chemical equilibrium in the reaction H2(g) + I2(g) 2HI(g) can be attained from either direction

Hydrogen and iodine, from both ends

- starting with equal initial concentrations of H2 and I2, the reaction proceeds forward and their concentrations decrease while that of HI increases, until all become constant at equilibrium
- we can also start with HI alone and let the reaction proceed in the reverse direction, when the concentration of HI decreases and those of H2 and I2 increase until all become constant
- if the total numbers of H and I atoms are the same in a given volume, the same equilibrium mixture is obtained whether we start from pure reactants or pure product

The deuterium experiment

- when a mixture of H2, N2 and NH3 is mixed with one of D2, N2 and ND3 after each has reached equilibrium, and the combination is left for a while, the concentration of ammonia is found to be just the same as before
- however a mass spectrometer finds ammonia and all its deuterium containing forms, NH3, NH2D, NHD2 and ND3, together with dihydrogen and its deutrated forms, H2, HD and D2, so the scrambling of H and D atoms must result from a continuation of the forward and reverse reactions, since if the reaction had simply stopped at equilibrium there would have been no mixing of isotopes
- so chemical reactions reach a state of dynamic equilibrium in which the rates of forward and reverse reactions are equal and there is no net change in composition, and equilibrium can be attained from both sides, whether we start from N2 and H2 and get NH3 or start from NH3 and decompose it

6.3 Law of chemical equilibrium and equilibrium constant

The questions to answer

- a mixture of reactants and products in the equilibrium state is called an equilibrium mixture, and the questions to address are what the relationship is between the concentrations of reactants and products in it, how we can determine equilibrium concentrations from initial ones, and what factors can be exploited to alter its composition

Six experiments, four started from H2 and I2 and two from HI alone

For each one the initial concentrations and then the equilibrium ones

The equilibrium figures agreeing whichever end the run began at

Figure 6.F Initial and Equilibrium Concentrations of H2, I2 and HI

Guldberg and Waage

- the question of what alters the composition of an equilibrium mixture is particularly important when choosing conditions for the synthesis of industrial chemicals such as H2, NH3 and CaO
- for a general reversible reaction of A and B giving C and D, the Norwegian chemists Cato Maximillian Guldberg and Peter Waage proposed in 1864, on the basis of experimental studies of many reversible reactions, that the concentrations in an equilibrium mixture are related by an equilibrium equation, in which Kc is the equilibrium constant and the expression on the right is called the equilibrium constant expression
- the equilibrium equation is also known as the law of mass action, because in the early days of chemistry concentration was called active mass

The six experiments

- taking the reaction between gaseous H2 and I2 in a sealed vessel at 731 K, six sets of experiments with varying initial conditions were performed, the first four starting with only gaseous H2 and I2 and the last two with only HI
- the first four were performed with different concentrations of H2 or I2, and with time the intensity of the purple colour remained constant and equilibrium was attained, while for the last two equilibrium was attained from the opposite direction
- it is evident from the first four that the number of moles of dihydrogen reacted equals the number of moles of iodine reacted and equals half the number of moles of HI formed, while the last two indicate that the equilibrium concentrations of H2 and I2 are equal
- to establish a relationship between the concentrations of reactants and products several combinations can be tried, starting with the simple expression of the HI concentration divided by the product of the H2 and I2 concentrations

The simple ratio of HI to H2 times I2 tried first, and coming out far from constant

The same ratio with the HI concentration squared instead

That one holding the same value across all six experiments

Figure 6.G Expression Involving the Equilibrium Concentration of Reactants H2(g) + I2(g) 2HI(g)

Squaring the HI term

- the simple expression is far from constant, however taking the square of the HI concentration over the product of the H2 and I2 concentrations gives a constant value in all six cases
- in that expression the powers of the concentrations for reactants and products are the stoichiometric coefficients in the equation for the chemical reaction
- generally the subscript for equilibrium is omitted from the concentration terms, since it is taken for granted that the concentrations in the expression are equilibrium values, and the subscript c indicates that Kc is expressed in concentrations of mol L-1

The equilibrium law

- at a given temperature, the product of the concentrations of the reaction products raised to their respective stoichiometric coefficients, divided by the product of the concentrations of the reactants raised to their individual stoichiometric coefficients, has a constant value, and this is known as the equilibrium law or law of chemical equilibrium
- so for a general reaction of aA plus bB giving cC plus dD, the equilibrium constant is the concentration of C to the power c times that of D to the power d, over the concentration of A to the power a times that of B to the power b
- molar concentrations are indicated by enclosing the species in square brackets, and while writing the expression the symbols for phases are generally ignored

Reversing the reaction

- for the reaction of H2 with I2 giving 2HI the equilibrium constant is the square of the HI concentration over the product of the other two, call it x
- for the reverse reaction at the same temperature the constant is the product of the H2 and I2 concentrations over the square of the HI concentration, which is 1 over x
- so the equilibrium constant of a reversed reaction is the reciprocal of the equilibrium constant of the forward one

The general reaction with its constant Kc

The same reaction reversed, whose constant is 1 over Kc

The same reaction with every coefficient multiplied by n, whose constant is Kc to the power n

Figure 6.H Relations between Equilibrium Constants for a General Reaction and its Multiples.

Multiplying the equation through

- if every coefficient in the equation is multiplied by n, the equilibrium constant for the new equation is Kc to the power n
- because the constants for a reaction and its reverse have different numerical values, it is important to specify the form of the balanced chemical equation when quoting the value of an equilibrium constant

Problems 6.1 and 6.2

- Problem 6.1 gives the equilibrium concentrations for the formation of NH3 from N2 and H2 at 500 K as 1.5 × 10-2 M, 3.0 × 10-2 M and 1.2 × 10-2 M respectively, and the equilibrium constant works out to 0.355 × 103, that is 3.55 × 102
- Problem 6.2 gives concentrations at equilibrium of N2 at 3.0 × 10-3 M, O2 at 4.2 × 10-3 M and NO at 2.8 × 10-3 M in a sealed vessel at 800 K, and for N2 plus O2 giving 2NO the constant is the square of the NO concentration over the product of the other two, which comes to 0.622

6.4 Homogeneous equilibria, and 6.4.1 equilibrium constant in gaseous systems

One phase throughout

- in a homogeneous system all the reactants and products are in the same phase, as in the gaseous reaction of N2 with H2 giving NH3, and likewise the hydrolysis of ethyl acetate and the reaction of the ferric ion with thiocyanate are homogeneous in solution phase
- so far equilibrium constants have been expressed in terms of molar concentration, with the symbol Kc, however for reactions involving gases it is usually more convenient to express the constant in terms of partial pressure
- from the ideal gas equation pV = nRT, the pressure equals n over V times RT, where p is the pressure in Pa, n the number of moles, V the volume in m3 and T the temperature in kelvin

A handful of reactions, each with its Kp at a stated temperature

The temperature has to be given, since the constant changes with it

Pressures in bar, because the standard state for pressure is 1 bar

Figure 6.I Equilibrium Constants, Kp for a Few Selected Reactions

Pressure in place of concentration

- n over V is the concentration expressed in mol per m3, so if the concentration is in mol per L and the pressure in bar then p equals cRT, that is the pressure of a gas equals its concentration times RT, with R at 0.0831 bar litre per mol per K
- at constant temperature the pressure of a gas is proportional to its concentration, so for the reaction of H2 with I2 giving 2HI the constant can be written either in concentrations or in partial pressures, and in this example the two come out equal
- that is not always the case, since for N2 plus 3H2 giving 2NH3 the pressure form differs from the concentration form by a factor of RT to the power minus 2

The general relation

- for a general reaction of aA plus bB giving cC plus dD, Kp equals Kc times RT to the power Δn, where Δn is the number of moles of gaseous products minus the number of moles of gaseous reactants in the balanced chemical equation
- it is necessary that while calculating the value of Kp the pressure should be expressed in bar, because the standard state for pressure is 1 bar
- one pascal is one newton per square metre, and one bar is 105 pascal

Problems 6.3 and 6.4

- Problem 6.3 has PCl5, PCl3 and Cl2 at equilibrium at 500 K with concentrations 1.59 M PCl3, 1.59 M Cl2 and 1.41 M PCl5, and Kc is the product of the two product concentrations over that of PCl5, which comes to 1.79
- Problem 6.4 gives Kc as 4.24 at 800 K for CO plus H2O giving CO2 plus H2, with only CO and H2O present initially at 0.10 M each, and asks for the equilibrium concentrations
- letting x mol per litre of each product be formed, the equilibrium concentrations are 0.1 minus x for each reactant and x for each product, so Kc equals x squared over the square of 0.1 minus x, which rearranges to the quadratic 3.24x squared minus 0.848x plus 0.0424 = 0
- solving gives 0.067 and 0.194, and the second must be neglected because it would give a reactant concentration greater than the initial one, so CO2 and H2 are each 0.067 M while CO and H2O are each 0.1 minus 0.067, that is 0.033 M

Problem 6.5

- for 2NOCl giving 2NO plus Cl2 the value of Kc is 3.75 × 10-6 at 1069 K, and the question asks for Kp at that temperature
- since Kp equals Kc times RT to the power Δn, and here Δn is 2 plus 1 minus 2, that is 1, Kp is 3.75 × 10-6 times 0.0831 times 1069, which gives 0.033

6.5 Heterogeneous equilibria

More than one phase

- equilibrium in a system having more than one phase is called heterogeneous equilibrium, and the equilibrium between water vapour and liquid water in a closed container is an example, there being a gas phase and a liquid phase
- in the same way the equilibrium between a solid and its saturated solution, as with calcium hydroxide and its ions in water, is a heterogeneous equilibrium
- heterogeneous equilibria often involve pure solids or liquids, and the equilibrium expression can be simplified on that account

Calcium carbonate decomposing

- in the thermal dissociation of calcium carbonate to calcium oxide and carbon dioxide, the concentrations of the two solids are both constant, so the modified equilibrium constant is simply the concentration of CO2, or in pressure terms its partial pressure
- this shows that at a particular temperature there is a constant concentration or pressure of CO2 in equilibrium with the two solids, and experimentally at 1100 K that pressure is 2.0 × 105 Pa, so the equilibrium constant at 1100 K is 2.00
- similarly in the equilibrium between nickel, carbon monoxide and nickel carbonyl, used in the purification of nickel, the solid nickel does not appear and the constant is the concentration of the carbonyl over the fourth power of the CO concentration

What must still be present

- for the existence of a heterogeneous equilibrium the pure solids or liquids must also be present, however small the amount may be, but their concentrations or partial pressures do not appear in the expression for the equilibrium constant
- so in the reaction of solid silver oxide with nitric acid to give silver nitrate and water, the constant carries only the two aqueous species

Units of the equilibrium constant

- the value of Kc is calculated by substituting the concentration terms in mol per L, and for Kp the partial pressure is substituted in Pa, kPa, bar or atm, which results in units based on molarity or pressure unless the exponents of the numerator and denominator are the same
- so for H2 plus I2 giving 2HI both Kc and Kp have no unit, whereas for N2O4 giving 2NO2 Kc has the unit mol per L and Kp the unit bar
- equilibrium constants can also be expressed as dimensionless quantities if the standard state of the reactants and products is specified

A worked equilibrium in partial pressures

- with a starting pressure of 0.48 bar and x consumed, the equilibrium partial pressures are 0.48 minus x for one species and 2x for the other, and Kp is the square of 2x over 0.48 minus x, set equal to 3
- that rearranges to the quadratic 4x squared plus 3x minus 1.44 = 0, whose solution, taking the root that is not negative, is 2.66 over 8, that is 0.33
- so the equilibrium partial pressures are 2x, that is 0.66 bar, and 0.48 minus x, that is 0.15 bar

6.6 Applications of equilibrium constants, and 6.6.1 predicting the extent

What the constant is good for

- the equilibrium constant for a reaction is related to that of the corresponding reaction whose equation is obtained by multiplying or dividing the original equation by a small integer
- the applications to consider are predicting the extent of a reaction from the magnitude of the constant, predicting the direction of the reaction, and calculating equilibrium concentrations

Reading the magnitude

- the numerical value of the equilibrium constant indicates the extent of the reaction, but it is important to note that it gives no information about the rate at which equilibrium is reached
- the magnitude of Kc or Kp is directly proportional to the concentrations of the products, which appear in the numerator, and inversely proportional to those of the reactants, which appear in the denominator, so a high value of K suggests a high concentration of products
- if Kc is greater than 103 the products predominate, that is the reaction proceeds nearly to completion, as with H2 and O2 at 500 K where Kc is 2.4 × 1047, H2 with Cl2 at 300 K where it is 4.0 × 1031, and H2 with Br2 at 300 K where it is 5.4 × 1018
- if Kc is less than 10-3 the reactants predominate, that is the reaction proceeds rarely

A scale of Kc running from very small to very large

Reactants alone at one end, products alone at the other

The middle band, 10-3 to 103, where appreciable amounts of both are present

Figure 6.J Dependence of extent of reaction on Kc

Qc smaller than Kc, so the net reaction runs left to right

Qc larger than Kc, so it runs right to left

Qc equal to Kc, so the mixture is already at equilibrium

Figure 6.K Predicting the direction of the reaction

The three bands of Kc

- the decomposition of H2O into H2 and O2 at 500 K has a very small equilibrium constant of 4.1 × 10-48, and N2 with O2 giving 2NO at 298 K has 4.8 × 10-31
- if Kc is in the range 10-3 to 103, appreciable concentrations of both reactants and products are present, as with H2 and I2 giving HI where Kc is 57.0 at 700 K, and the gas phase decomposition of N2O4 to NO2 where Kc is 4.64 × 10-3 at 25 degrees C, so equilibrium mixtures contain appreciable concentrations of both

6.6.2 Predicting the direction of the reaction

The reaction quotient

- the equilibrium constant helps in predicting the direction in which a given reaction will proceed at any stage, and for this purpose we calculate the reaction quotient Q, written Qc with molar concentrations and Qp with partial pressures
- it is defined in the same way as the equilibrium constant except that the concentrations in it are not necessarily equilibrium values, so for a general reaction Qc is the same expression as Kc but with the concentrations measured at some arbitrary time
- taking H2 with I2 at 700 K where Kc is 57.0, if the concentrations at some time are 0.10 M, 0.20 M and 0.40 M then Qc is the square of 0.40 over 0.10 times 0.20, that is 8.0, which does not equal Kc, so the mixture is not at equilibrium and more H2 and I2 will react to form HI until Qc equals Kc

Problem 6.7, reading the direction

- with all three species at 3 × 10-4 M, the reaction quotient works out to 1
- since Qc is greater than Kc, the reaction will proceed in the reverse direction

Problem 6.8, dinitrogen tetroxide at 400 K

- 13.8 g of N2O4 in a 1 L vessel at 400 K is 0.15 mol, so from pV = nRT with R at 0.083 bar L mol-1 K-1 the initial pressure is 4.98 bar
- with x consumed the equilibrium pressures are 4.98 minus x for N2O4 and 2x for NO2, and since the total at equilibrium is 9.15 bar, x is 9.15 minus 4.98, that is 4.17 bar
- so the partial pressures are 0.81 bar and 8.34 bar, giving Kp as the square of 8.34 over 0.81, that is 85.87, and since Kp equals Kc times RT to the power Δn with Δn of 1, Kc comes to 2.6

6.6.3 Calculating equilibrium concentrations

The three steps

- step 1 is to write the balanced equation for the reaction
- step 2 is to make a table under it listing for each substance the initial concentration, the change in concentration on going to equilibrium, and the equilibrium concentration, defining x as the concentration of one of the substances that reacts and using the stoichiometry to express the others in terms of x
- step 3 is to substitute the equilibrium concentrations into the equilibrium equation and solve for x, choosing the chemically sensible root if it is a quadratic

Problem 6.9 finished

- letting x mol per litre of PCl5 dissociate, the equilibrium concentrations are 3 minus x for PCl5 and x for each product, so Kc equals x squared over 3 minus x, set equal to 1.8
- that rearranges to x squared plus 1.8x minus 5.4 = 0, whose positive root is 1.59
- so PCl5 is 3.0 minus 1.59, that is 1.41 M, while PCl3 and Cl2 are each 1.59 M

6.7 Relationship between K, Q and Gibbs energy G

Reading spontaneity from the constant

- the value of Kc does not depend on the rate of the reaction, however it is directly related to the thermodynamics of the reaction and in particular to the change in Gibbs energy
- if the standard Gibbs energy change is negative then its negative over RT is positive and the exponential is greater than 1, making K greater than 1, which implies a spontaneous reaction proceeding forward to the extent that products predominate
- if it is positive then the exponential is less than 1, so K is less than 1, which implies a non-spontaneous reaction proceeding forward to such a small degree that only a very minute quantity of product is formed

Problems 6.10 and 6.11

- Problem 6.10 gives the standard Gibbs energy change for the phosphorylation of glucose in glycolysis as 13.8 kJ per mol and asks for Kc at 298 K, and since it equals minus RT ln Kc, ln Kc is minus 5.569 and Kc is 3.81 × 10-3
- Problem 6.11 gives the hydrolysis of sucrose to glucose and fructose with an equilibrium constant of 2 × 1013 at 300 K and asks for the standard Gibbs energy change, which comes to minus 7.64 × 104 J mol-1

Concentrations level at equilibrium, then hydrogen added in a step

Hydrogen and iodine falling away again while HI climbs

All three levelling at a new set of values

Figure 6.L Effect of addition of H2 on change of concentration for the reactants and products in the reaction, H2(g) + I2 (g) 2HI(g)

6.8 Factors affecting equilibria

Why the conditions are chosen as they are

- one of the principal goals of chemical synthesis is to maximise the conversion of the reactants to products while minimising the expenditure of energy, which implies maximum yield at mild temperature and pressure, and if that does not happen the experimental conditions need to be adjusted
- in the Haber process for the synthesis of ammonia the choice of conditions is of real economic importance, annual world production being about a hundred million tonnes, primarily for use as fertilisers
- Kc is independent of the initial concentrations, but if a system at equilibrium is subjected to a change in the concentration of one or more of the reacting substances then it is no longer at equilibrium, and a net reaction takes place in some direction until it returns, while a change in temperature or pressure may also alter the equilibrium

The principle itself, and concentration change

- Le Chatelier's principle states that a change in any of the factors that determine the equilibrium conditions of a system will cause the system to change in such a manner as to reduce or counteract the effect of the change, and this is applicable to all physical and chemical equilibria
- the concentration stress of an added reactant or product is relieved by net reaction in the direction that consumes the added substance
- the concentration stress of a removed reactant or product is relieved by net reaction in the direction that replenishes the removed substance

Adding a reactant, removing a product

- addition of hydrogen at equilibrium makes Qc less than Kc, so to attain equilibrium again the reaction moves in the forward direction, and removal of a product likewise boosts the forward reaction and increases the concentration of the products
- this has great commercial application where the product is a gas or a volatile substance, so in the manufacture of ammonia the ammonia is liquified and removed from the reaction mixture to keep the reaction moving forward, and in the large scale production of CaO from CaCO3 the constant removal of CO2 from the kiln drives the reaction to completion
- continuous removal of a product maintains Qc at a value less than Kc, so the reaction continues to move in the forward direction

An experiment with the thiocyanate complex

- a reddish colour appears on adding two drops of 0.002 M potassium thiocynate solution to 1 mL of 0.2 M iron(III) nitrate solution, due to the formation of the deep red complex, and the intensity becomes constant on attaining equilibrium
- the equilibrium can be shifted in either direction depending on whether a reactant or a product is added, and it can be shifted the opposite way by adding reagents that remove the ferric or the thiocyanate ions, for example oxalic acid reacts with ferric ions to form a stable complex, decreasing the concentration of free ferric ion
- in accordance with Le Chatelier's principle the concentration stress of the removed ferric ion is relieved by dissociation of the red complex to replenish it, and because the concentration of the complex decreases, the intensity of the red colour decreases
- addition of aqueous mercuric chloride also decreases the red colour, because the mercuric ion reacts with the thiocyanate ion

Two sealed tubes of the same brown gas mixture, one warm and one cold

The warm one darker, since heat drives the equilibrium towards NO2

The cold one paler, since cooling drives it towards the colourless N2O4

Figure 6.M Effect of temperature on equilibrium for the reaction, 2NO2 (g) N2O4 (g)

Effect of pressure, and of an inert gas

- when the volume is halved the partial pressure and the concentration of every gas are doubled, so replacing each equilibrium concentration by double its value gives a reaction quotient that can be compared with Kc to see which way the reaction moves
- in the reaction of solid carbon with CO2 to give 2CO, increasing the pressure drives the reaction in the reverse direction, because the number of moles of gas increases in the forward direction
- if the volume is kept constant and an inert gas such as argon is added, which does not take part in the reaction, the equilibrium remains undisturbed, because the addition at constant volume does not change the partial pressures or molar concentrations of the substances involved, and the reaction quotient changes only if the added gas is itself a reactant or product

Effect of temperature change

- whenever an equilibrium is disturbed by a change in concentration, pressure or volume, the composition changes because Qc no longer equals Kc, however when a change in temperature occurs it is the value of Kc itself that changes
- the temperature dependence of the equilibrium constant depends on the sign of ΔH, so the constant for an exothermic reaction decreases as the temperature increases, while that for an endothermic reaction increases
- the production of ammonia from N2 and H2 is exothermic at minus 92.38 kJ mol-1, so by Le Chatelier's principle raising the temperature works against the yield

Temperature seen in colour

- raising the temperature favours the formation of NO2, and so the brown colour intensifies
- the effect can also be seen in an endothermic reaction, that of the hexaaqua cobalt ion with chloride to give the tetrachloro complex, where at room temperature the equilibrium mixture is blue due to the tetrachloro complex, while cooling in a freesing mixture turns it pink due to the aqua complex

6.8.5 Effect of a catalyst

What a catalyst does and does not do

- a catalyst increases the rate of a chemical reaction by making available a new low energy pathway for the conversion of reactants to products, increasing the rates of the forward and reverse reactions, which pass through the same transition state, and so it does not affect the equilibrium
- it lowers the activation energy for the forward and reverse reactions by exactly the same amount, so it does not affect the equilibrium composition of a reaction mixture, and it does not appear in the balanced chemical equation or in the equilibrium constant expression

Choosing the conditions for ammonia

- the formation of NH3 from dinitrogen and dihydrogen is highly exothermic and proceeds with a decrease in the total number of moles, so the equilibrium constant decreases with increase in temperature, and at low temperature the rate is slow and takes a long time to reach equilibrium whereas high temperatures give satisfactory rates but poor yields
- the German chemist Fritz Haber discovered that a catalyst consisting of iron catalyses the reaction to occur at a satisfactory rate at temperatures where the equilibrium concentration of NH3 is reasonably favourable, and since the number of moles formed is less than that of the reactants, the yield can be improved by increasing the pressure
- the optimum conditions for the synthesis of NH3 using a catalyst are around 500 degrees C and 200 atm

A crystal of sodium chloride meeting water

Polar water molecules turning their negative ends to Na+ and positive ends to Cl-

Each ion carried away wrapped in a shell of water, which is what stabilises it

Figure 6.N Dissolution of sodium chloride in water. Na+ and Cl– ions are stablised by their hydration with polar water molecules.

6.10 Acids, bases and salts

Ionic equilibrium

- in weak electrolytes an equilibrium is established between the ions and the unionized molecules, and this type of equilibrium involving ions in aqueous solution is called ionic equilibrium
- acids, bases and salts come under the category of electrolytes and may act as either strong or weak electrolytes

Where they are found

- hydrochloric acid present in the gastric juice is secreted by the lining of our stomach in a significant amount of 1.2 to 1.5 L per day and is essential for digestive processes, acetic acid is the main constituent of vinegar, lemon and orange juices contain citric and ascorbic acids, and tartaric acid is found in tamarind paste
- as most acids taste sour, the word acid has been derived from the latin acidus meaning sour, and acids turn blue litmus paper red and liberate dihydrogen on reacting with some metals, while bases turn red litmus paper blue, taste bitter and feel soapy, a common example being washing soda
- when acids and bases are mixed in the right proportion they react to give salts, such as sodium chloride, barium sulphate and sodium nitrate, and sodium chloride is an important component of our diet, formed by the reaction between hydrochloric acid and sodium hydroxide
- it exists in the solid state as a cluster of positively charged sodium ions and negatively charged chloride ions held together by electrostatic interactions, and since the electrostatic force between two charges is inversely proportional to the dielectric constant of the medium, water, with its very high dielectric constant, pulls them apart

Dissociation against ionization

- a strong electrolyte is completely ionized into its constituent ions while a weak one is only partially ionized, less than 5 per cent, the extent depending upon the strength of the bond and the extent of solvation of the ions produced
- dissociation refers to the separation in water of ions already existing as such in the solid state of the solute, as in sodium chloride, whereas ionization is a process in which a neutral molecule splits into charged ions in solution, and here the two terms are used interchangeably

6.10.1 Arrhenius concept of acids and bases

Hydrogen ions and hydroxyl ions

- according to Arrhenius theory, acids are substances that dissociate in water to give hydrogen ions and bases are substances that produce hydroxyl ions
- a bare proton is very reactive and cannot exist freely in aqueous solution, so it bonds to the oxygen atom of a solvent water molecule to give the trigonal pyramidal hydronium ion, and the two notations are used interchangeably to mean a hydrated proton, while the hydroxyl ion also exists in hydrated form
- the Arrhenius concept suffers from the limitation of being applicable only to aqueous solutions, and it does not account for the basicity of substances like ammonia which do not possess a hydroxyl group

6.10.2 The Brönsted-Lowry acids and bases

Proton donors and acceptors

- on this view an acid is a substance that donates a proton and a base is one that accepts a proton, which frees the definitions from any dependence on water

Svante Arrhenius

- Arrhenius was born near Uppsala, Sweden, and presented his thesis on the conductivities of electrolyte solutions to the University of Uppsala in 1884, then travelled extensively for five years visiting research centres in Europe
- in 1895 he was appointed professor of physics at the newly formed University of Stockholm, serving as its rector from 1897 to 1902, and from 1905 until his death he was Director of physical chemistry at the Nobel Institute in Stockholm, continuing to work for many years on electrolytic solutions, and in 1899 he discussed the temperature dependence of reaction rates on the basis of an equation now usually known as the Arrhenius equation
- he worked in a variety of fields and made important contributions to immunochemistry, cosmology, the origin of life and the causes of the ice age, being the first to discuss the green house effect and to call it by that name, and he received the Nobel Prize in Chemistry in 1903 for his theory of electrolytic dissociation

Conjugate acid-base pairs

- in the reverse of the water and ammonia reaction a proton is transferred from the ammonium ion to the hydroxyl ion, so the ammonium ion acts as a Bronsted acid while the hydroxyl ion acts as a Brönsted base
- an acid-base pair that differs only by one proton is called a conjugate acid-base pair, so the hydroxyl ion is the conjugate base of the acid water and the ammonium ion is the conjugate acid of the base ammonia
- if a Brönsted acid is a strong acid then its conjugate base is a weak base and the other way round, and it may be noted that a conjugate acid has one extra proton and each conjugate base one less proton
- in the ionization of hydrochloric acid in water, HCl acts as an acid by donating a proton to the water molecule, which acts as a base, whereas with ammonia water acts as an acid by donating a proton

6.10.3 Lewis acids and bases

Electron pairs instead of protons

- G.N. Lewis in 1923 defined an acid as a species which accepts an electron pair and a base as one which donates an electron pair
- as far as bases are concerned there is not much difference between the Brönsted-Lowry and Lewis concepts, since the base provides a lone pair in both cases, however in the Lewis concept many acids do not have a proton
- a typical example is the reaction of the electron deficient species BF3 with NH3, where BF3 has no proton but still acts as an acid by accepting the lone pair of ammonia, and electron deficient species like AlCl3, Co3+ and Mg2+ can act as Lewis acids while species like H2O, NH3 and OH- can act as Lewis bases

Problem 6.15

- the hydroxyl ion is a Lewis base as it can donate an electron lone pair, and the fluoride ion acts as a Lewis base as it can donate any one of its four electron lone pairs
- a proton is a Lewis acid as it can accept a lone pair from bases like the hydroxyl ion and the fluoride ion, and BCl3 acts as a Lewis acid as it can accept a lone pair from species like ammonia or amine molecules

6.11 Ionization of acids and bases

Strong acids

- the Arrhenius concept becomes useful for the ionization of acids and bases, since most ionizations in chemical and biological systems occur in aqueous medium
- strong acids like perchloric acid, hydrochloric acid, hydrobromic acid, hydroiodic acid, nitric acid and sulphuric acid are termed strong because they are almost completely dissociated into their constituent ions in an aqueous medium

Strong acids have weak conjugate bases

- a stronger acid donates a proton to a stronger base, giving a weaker acid and a weaker base, so as a strong acid dissociates completely in water the base formed is very weak
- the strong acids perchloric, hydrochloric, hydrobromic, hydroiodic, nitric and sulphuric give conjugate base ions which are much weaker bases than water, and similarly a very strong base gives a very weak conjugate acid
- a weak acid is only partially dissociated so the solution mainly contains undissociated molecules, typical weak acids being nitrous acid, hydrofluoric acid and acetic acid, and weak acids have very strong conjugate bases, the amide, oxide and hydride ions being very good proton acceptors and much stronger bases than water

Indicators

- certain water soluble organic compounds like phenolphthalein and bromothymol blue behave as weak acids and exhibit different colours in their acid and conjugate base forms
- such compounds are useful as indicators in acid-base titrations and for finding out the hydrogen ion concentration

6.11.1 The ionization constant of water and its ionic product

Water as both acid and base

- some substances like water are unique in their ability to act both as an acid and a base, so in the presence of an acid water accepts a proton and acts as the base, while in the presence of a base it acts as an acid by donating a proton
- in pure water one water molecule donates a proton to another, which is the self-ionization that fixes the ionic product

What pH is

- in dilute solutions below 0.01 M the activity of the hydrogen ion is equal in magnitude to the molarity, and activity has no units, being defined as the concentration divided by mol L-1, so pH is minus the logarithm of that quantity
- an acidic solution of HCl at 10-2 M has a pH of 2, a basic solution of NaOH with a hydroxide concentration of 10-4 M and a hydronium concentration of 10-10 M has a pH of 10, and at 25 degrees C pure water has a hydrogen ion concentration of 10-7 M so its pH is 7
- acidic solutions have a hydrogen ion concentration greater than 10-7 M and basic solutions less, so an acidic solution has pH below 7, a basic solution above 7 and a neutral solution exactly 7

pKw, and what one unit of pH means

- taking negative logarithms of the ionic product of water at 298 K gives pKw equal to pH plus pOH equal to 14, and although Kw may change with temperature the variations in pH with temperature are so small that we often ignore them
- pKw is a very important quantity for aqueous solutions and controls the relative concentrations of hydrogen and hydroxyl ions, since their product is a constant
- as the pH scale is logarithmic, a change in pH by just one unit means a change in the hydrogen ion concentration by a factor of 10, and when that concentration changes by a factor of 100 the pH changes by 2 units, which is why the change in pH with temperature is often ignored

A very dilute acid, where water matters

- for a very dilute HCl solution the hydronium ion comes both from the ionization of the dissolved acid and from the ionization of water, so both sources must be considered
- writing x for the contribution from water gives the hydronium concentration as 10-8 plus x, and substituting into the ionic product gives the quadratic x squared plus 10-8 x minus 10-14 = 0
- solving gives a hydroxide concentration of 9.5 × 10-8, so pOH is 7.02 and pH is 6.98

6.11.3 Ionization constants of weak acids

Setting up the equilibrium

- consider a weak acid HX partially ionized in aqueous solution, with initial concentration c and no products, and let alpha be the extent of ionization
- the change is minus c alpha for the acid and plus c alpha for each product, so at equilibrium the concentrations are c minus c alpha, c alpha and c alpha
- the equilibrium constant is therefore c alpha squared over 1 minus alpha, and this is called the dissociation or ionization constant of the acid, which can also be written as the product of the hydrogen ion and X concentrations over the concentration of HX
- at a given temperature Ka is a measure of the strength of the acid, a larger value meaning a stronger acid

Table 6.A The Ionization Constants of Some Selected Weak Acids (at 298K)

Acid	Ionization Constant, Ka
Hydrofluoric Acid (HF)	3.5 × 10–4
Nitrous Acid (HNO2)	4.5 × 10–4
Formic Acid (HCOOH)	1.8 × 10–4
Niacin (C5H4NCOOH)	1.5 × 10–5
Acetic Acid (CH3COOH)	1.74 × 10–5
Benzoic Acid (C6H5COOH)	6.5 × 10–5
Hypochlorous Acid (HClO)	3.0 × 10–8
Hydrocyanic Acid (HCN)	4.9 × 10–10
Phenol (C6H5OH)	1.3 × 10–10

pKa, and the step-wise approach

- pKa is minus the logarithm of Ka, and knowing the ionization constant of an acid and its initial concentration it is possible to calculate the equilibrium concentration of all species, the degree of ionization and the pH
- step 1 identifies the species present before dissociation as Brönsted-Lowry acids or bases, and step 2 writes balanced equations for all possible reactions, including any species acting both as acid and base
- step 3 identifies the reaction with the higher Ka as the primary reaction and the other as subsidiary, and step 4 lists in tabular form, for each species in the primary reaction, the initial concentration, the change on proceeding to equilibrium in terms of the degree of ionization, and the equilibrium concentration
- step 5 substitutes the equilibrium concentrations into the equilibrium constant equation for the principal reaction and solves for the degree of ionization, step 6 calculates the concentrations of the species, and step 7 calculates the pH as minus the logarithm of the hydronium concentration

Problem 6.18, hydrofluoric acid

- two proton transfer reactions are possible, the ionization of HF with a Ka of 3.2×10^{-4} and the self-ionization of water with a Kw of 1.0×10^{-14}
- since Ka is very much larger than Kw, the ionization of HF is the primary reaction and the hydronium ion from water can be neglected

Problem 6.19, and percent dissociation

- from a pH of 4.50 the hydrogen ion concentration is 10 to the minus 4.50, that is 3.16 × 10-5, which is also the concentration of the anion
- since the equilibrium concentration of the undissociated acid is 0.1 minus that, which is very nearly 0.1, Ka is the square of 3.16 × 10-5 over 0.1, that is 1.0 × 10-8, so pKa is 8
- percent dissociation is another useful measure of the strength of a weak acid, being the dissociated concentration over the initial concentration times 100 per cent

Problem 6.20, hypochlorous acid

- for a 0.08 M solution of HOCl with an ionization constant of 2.5 × 10-5, letting x be the amount ionized gives equilibrium concentrations of 0.08 minus x for the acid and x for each ion
- so Ka is x squared over 0.08 minus x, and since x is much smaller than 0.08 the denominator is very nearly 0.08, giving x squared as 2.0 × 10-6 and x as 1.41 × 10-3
- so the hydrogen ion concentration is 1.41 × 10-3 M, from which the pH and the percent dissociation follow

Table 6.B The Values of the Ionization Constant of Some Weak Bases at 298 K

Base	Kb
Dimethylamine, (CH3)2NH	5.4 × 10–4
Triethylamine, (C2H5)3N	6.45 × 10–5
Ammonia, NH3 or NH4OH	1.77 × 10–5
Quinine, (A plant product)	1.10 × 10–6
Pyridine, C5H5N	1.77 × 10–9
Aniline, C6H5NH2	4.27 × 10–10
Urea, CO (NH2)2	1.3 × 10–14

6.11.5 Ionization of weak bases

Weak bases and pKb

- quinine, pyridine, aniline and nicotine all behave as very weak bases because of their very small Kb, while ammonia produces hydroxide ion in aqueous solution
- the pH scale for the hydrogen ion concentration has been extended so that pKb is minus the logarithm of Kb

Problem 6.21, hydrazine

- the pH of a 0.004 M hydrazine solution is 9.7, and the question asks for its ionization constant and pKb
- from the pH the hydrogen ion concentration is the antilog of minus 9.7, that is 1.67 × 10-10, and dividing the ionic product of water by that gives a hydroxide concentration of 5.98 × 10-5
- the concentration of the hydrazinium ion is the same, and since both are very small the undissociated base can be taken as 0.004 M, so Kb is the square of 5.98 × 10-5 over 0.004, that is 8.96 × 10-7, and pKb is 6.04

Problem 6.22, an ammonia and ammonium chloride mixture

- the question asks for the pH of a solution containing 0.2 M NH4Cl and 0.1 M NH3, with the pKb of ammonia solution given as 4.75
- the ionization constant of NH3 is the antilog of minus 4.75, that is 1.77 × 10-5 M
- setting up the table with initial concentrations of 0.10 for ammonia and 0.20 for the ammonium ion, and letting x be the change, the equilibrium expression can be solved for the hydroxide concentration and hence the pH

6.11.6 Relation between Ka and Kb

Adding reactions multiplies their constants

- the equilibrium constant for a net reaction obtained after adding two or more reactions equals the product of the equilibrium constants for the individual reactions
- similarly, for a conjugate acid-base pair, Ka times Kb equals Kw, so knowing one the other can be obtained, and it should be noted that a strong acid will have a weak conjugate base and the other way round

The same result from the base equilibrium

- starting from the base-dissociation equilibrium, Kb is the product of the concentrations of the conjugate acid and the hydroxide ion over the concentration of the base, the concentration of water being constant and so omitted and incorporated within the constant
- multiplying and dividing that expression by the hydrogen ion concentration separates it into the ionic product of water over Ka, so again Ka times Kb equals Kw
- taking negative logarithms of both sides relates the pK values of the conjugate acid and base, so pKa plus pKb equals pKw equals 14 at 298 K

Problem 6.23

- the question asks for the degree of ionization and the pH of a 0.05 M ammonia solution, taking the ionization constant from the table of weak bases, and also for the ionization constant of the conjugate acid of ammonia
- the ionization of ammonia in water gives the ammonium ion and the hydroxide ion, and the conjugate acid constant follows from Ka times Kb equals Kw

Each polyprotic acid with its successive ionization constants

Ka1 for the first proton, then Ka2, then Ka3 where there is one

Each one smaller than the one before it

Figure 6.0 The Ionization Constants of Some Common Polyprotic Acids (298K)

Why later constants are smaller

- Ka1 and Ka2 are the first and second ionization constants of a dibasic acid, and similarly tribasic acids like H3PO4 have three
- higher order ionization constants are smaller than the lower order ones, because it is more difficult to remove a positively charged proton from a negative ion due to electrostatic forces, as in removing a proton from uncharged H2CO3 compared with negatively charged HCO3-, or from doubly charged HPO4 compared with H2PO4-

What makes an acid strong

- in general, when the strength of the H-A bond decreases, that is the energy required to break it decreases, HA becomes a stronger acid, and when the H-A bond becomes more polar, that is the electronegativity difference increases and there is marked charge separation, cleavage of the bond becomes easier and the acidity increases
- while comparing elements in the same group, H-A bond strength is a more important factor than polarity, so as the size of A increases down the group the bond strength decreases and the acid strength increases, which is why acid strength runs HF, HCl, HBr, HI, and H2S is a stronger acid than H2O
- when comparing elements in the same row, bond polarity becomes the deciding factor, so as the electronegativity of A increases the acid strength increases, running CH4, NH3, H2O, HF

6.11.8 Common ion effect in the ionization of acids and bases

What the common ion effect is

- the common ion effect can be defined as a shift in equilibrium on adding a substance that provides more of an ionic species already present in the dissociation equilibrium
- so it is a phenomenon based on Le Chatelier's principle
- to evaluate the pH on adding 0.05 M acetate ion to 0.05 M acetic acid, the initial concentrations are 0.05 for the acid, 0 for the hydrogen ion and 0.05 for the acetate, and letting x be the extent of ionization the equilibrium concentrations are 0.05 minus x, x and 0.05 plus x
- since Ka is small for a very weak acid, x is much less than 0.05, so both 0.05 plus x and 0.05 minus x are very nearly 0.05, and the expression reduces to x equal to Ka, that is 1.8 × 10-5 M, giving a pH of 4.74

Problem 6.24, ammonia part-neutralised

- the question asks for the pH of a 0.10 M ammonia solution, and then the pH after 50.0 mL of it is treated with 25.0 mL of 0.10 M HCl, with Kb given as 1.77 × 10-5
- before neutralization the ammonium and hydroxide concentrations are both x while the ammonia is 0.10 minus x, very nearly 0.10, so x squared over 0.10 equals Kb, giving x as 1.33 × 10-3, which is the hydroxide concentration
- the hydrogen ion concentration then follows from the ionic product of water divided by that

6.12 Hydrolysis of salts and the pH of their solutions

Which salts hydrolyse

- ions formed on ionization of salts either exist as hydrated ions in aqueous solution or interact with water to reform the corresponding acids or bases depending on the nature of the salt, and this latter process is called hydrolysis, which affects the pH of the solution
- the cations of strong bases, such as Na+, K+, Ca2+ and Ba2+, and the anions of strong acids, such as Cl-, Br-, NO3- and ClO4-, simply get hydrated but do not hydrolyse, so solutions of salts formed from strong acids and bases are neutral with a pH of 7
- the other categories do undergo hydrolysis, namely salts of a weak acid and strong base such as sodium acetate, salts of a strong acid and weak base such as ammonium chloride, and salts of a weak acid and weak base such as ammonium acetate

Sodium acetate as the worked case

- sodium acetate, being a salt of the weak acid acetic acid and the strong base sodium hydroxide, gets completely ionised in aqueous solution
- the acetate ion then undergoes hydrolysis in water to give acetic acid and hydroxide ions, and since acetic acid is a weak acid it remains mainly unionised in solution
- this results in an increase of the hydroxide ion concentration, making the solution alkaline, so the pH of such a solution is more than 7, while ammonium chloride, formed from a weak base, runs the other way

Buffers

- solutions of known pH can be prepared from a knowledge of the pKa of an acid or the pKb of a base, and by controlling the ratio of the salt to the acid or the salt to the base
- a mixture of acetic acid and sodium acetate acts as a buffer solution around pH 4.75, and a mixture of ammonium chloride and ammonium hydroxide acts as a buffer around pH 9.25

6.12.1 Designing buffer solution

The Henderson-Hasselbalch equation

- to prepare a buffer of acidic pH we use a weak acid and its salt formed with a strong base, and we develop the equation relating the pH, the equilibrium constant of the weak acid, and the ratio of the concentrations of the weak acid and its conjugate base
- for the general case where the weak acid HA ionises in water, writing the equilibrium expression, rearranging it and taking logarithms of both sides gives what is known as the Henderson-Hasselbalch equation
- the quantity in it is the ratio of the concentration of the conjugate base of the acid to that of the acid present in the mixture, and since the acid is weak it ionises to a very little extent, so its concentration is negligibly different from the concentration of acid taken to form the buffer

A basic buffer

- for a buffer solution formed by ammonium chloride and ammonium hydroxide the equation becomes 9.25 plus the logarithm of the ratio of the concentration of the base to that of its conjugate acid
- the pH of the buffer solution is not affected by dilution, because the ratio under the logarithmic term remains unchanged

6.13 Solubility equilibria of sparingly soluble salts

Why some salts dissolve and others do not

- the solubility of ionic solids in water varies a great deal, some like calcium chloride being so soluble that they are hygroscopic and even absorb water vapour from the atmosphere, while others such as lithium fluoride have so little solubility that they are commonly termed insoluble
- the solubility depends on a number of factors, the most important being the lattice enthalpy of the salt and the solvation enthalpy of the ions in solution, since for a salt to dissolve the strong forces of attraction between its ions must be overcome by the ion-solvent interactions
- the solvation enthalpy of ions is always negative, that is energy is released in the process of solvation, and the amount depends on the nature of the solvent, so in a non-polar covalent solvent the solvation enthalpy is small and hence not sufficient to overcome the lattice enthalpy

Common ionic salts, each with its solubility product at 298 K

The values spanning many orders of magnitude

The smaller the value, the less of the salt a saturated solution can hold

Figure 6.P The Solubility Product Constants, Ksp of Some Common Ionic Salts at 298K.

Solubility product from molar solubility

- a solid salt of general formula MxXy with molar solubility S in equilibrium with its saturated solution dissolves to give x of the cation and y of the anion, where the charges balance
- its solubility product constant is the cation concentration to the power x times the anion concentration to the power y, which comes to x to the x times y to the y times S to the power x plus y, and rearranging gives the solubility in terms of the solubility product
- the term Ksp is replaced by Qsp when the concentration of one or more species is not the equilibrium concentration, so under equilibrium the two are equal but otherwise their comparison gives the direction of precipitation or dissolution

Problem 6.26

- the question asks for the solubility of A2X3 in pure water, assuming neither ion reacts with water, given a solubility product of 1.1 × 10-23
- writing S for the solubility, the cation concentration is 2S and the anion concentration 3S, so the solubility product is the square of 2S times the cube of 3S, that is 108 S to the fifth
- setting that equal to 1.1 × 10-23 gives S to the fifth as 1 × 10-25, so S is 1.0 × 10-5 mol per L

Problem 6.27, comparing two sparingly soluble salts

- for silver cyanide the solubility product is the product of the two ion concentrations at 6 × 10-17, and for nickel hydroxide it is the nickel concentration times the square of the hydroxide concentration at 2 × 10-15
- writing S1 for the silver concentration gives S1 squared equal to 6 × 10-17, so S1 is 7.8 × 10-9, and writing S2 for the nickel concentration gives S2 times the square of 2S2 equal to 2 × 10-15, so S2 is 0.58 × 10-4
- so nickel hydroxide is more soluble than silver cyanide

6.13.2 Common ion effect on solubility of ionic salts

Pushing a salt out of solution

- it is expected from Le Chatelier's principle that if we increase the concentration of any one of the ions it should combine with the ion of opposite charge and some of the salt will be precipitated until the solubility product and the reaction quotient are equal again, while if the concentration of one ion is decreased more salt will dissolve
- this applies even to soluble salts like sodium chloride, except that because of the higher concentrations we use activities instead of molarities in the expression
- so if we take a saturated solution of sodium chloride and pass HCl gas through it, sodium chloride is precipitated by the increased activity of chloride ion from the dissociation of HCl, and the sodium chloride thus obtained is of very high purity, free of impurities like sodium and magnesium sulphates
- the common ion effect is also used for the almost complete precipitation of a particular ion as its sparingly soluble salt

Summary

Equilibrium and the equilibrium constant

- when the number of molecules leaving the liquid to vapour equals the number returning, equilibrium is said to be attained and is dynamic in nature, and equilibrium can be established for both physical and chemical processes, at which stage the rates of the forward and reverse reactions are equal
- the equilibrium constant Kc is expressed as the concentration of products divided by that of reactants, each term raised to its stoichiometric coefficient, and it has a constant value at a fixed temperature, at which stage all the macroscopic properties such as concentration and pressure become constant
- for a gaseous reaction the equilibrium constant is expressed as Kp, written by replacing the concentration terms with partial pressures, and the direction of a reaction can be predicted by the reaction quotient Qc, which equals Kc at equilibrium

Le Chatelier, and ionic equilibrium

- Le Chatelier's principle states that a change in any factor such as temperature, pressure or concentration will cause the equilibrium to shift in such a direction as to reduce or counteract the effect of the change, and it can be used to study the effect of temperature, concentration, pressure, catalyst and inert gases on the direction of equilibrium and to control the yield
- acids, bases and salts in aqueous solution reach ionic equilibrium, described by the Arrhenius, Brönsted-Lowry and Lewis concepts, and quantified by the ionization constants Ka and Kb, which are related through the ionic product of water
- the pH scale expresses the hydrogen ion concentration logarithmically, buffer solutions hold a pH nearly constant against dilution, and for sparingly soluble salts the solubility product governs how much will dissolve and when precipitation occurs

Suggested activities, and exercises

Suggested activities for students

- use pH paper to determine the pH of fresh juices of various vegetables and fruits, soft drinks, body fluids and water samples, and also of different salt solutions, from which you may determine whether they are formed from strong or weak acids and bases
- prepare some buffer solutions by mixing solutions of sodium acetate and acetic acid and determine their pH using pH paper, and use different indicators to observe their colours in solutions of varying pH
- perform some acid-base titrations using indicators, observe the common ion effect on the solubility of sparingly soluble salts, and if a pH meter is available measure the pH with it and compare the results with those from pH paper

Exercises 6.1 to 6.5

- 6.1 A liquid is in equilibrium with its vapour in a sealed container at a fixed temperature. The volume of the container is suddenly increased. a) What is the initial effect of the change on vapour pressure? b) How do rates of evaporation and condensation change initially? c) What happens when equilibrium is restored finally and what will be the final vapour pressure?
- 6.2 What is Kc for the following equilibrium when the equilibrium concentration of each substance is: [SO2]= 0.60M, [O2] = 0.82M and [SO3] = 1.90M ? 2SO2(g) + O2(g) 2SO3(g)
- 6.3 At a certain temperature and total pressure of 105 Pa, iodine vapour contains 40% by volume of I atoms. I2 (g) 2I (g). Calculate Kp for the equilibrium.
- 6.4 Write the expression for the equilibrium constant, Kc for each of the following reactions: (i) 2NOCl (g) 2NO (g) + Cl2 (g) (ii) 2Cu(NO3)2 (s) 2CuO (s) + 4NO2 (g) + O2 (g) (iii) CH3COOC2H5(aq) + H2O(l) CH3COOH (aq) + C2H5OH (aq) (iv) Fe3+ (aq) + 3OH– (aq) Fe(OH)3 (s) (v) I2 (s) + 5F2 2IF5
- 6.5 Find out the value of Kc for each of the following equilibria from the value of Kp: (i) 2NOCl (g) 2NO (g) + Cl2 (g); Kp= 1.8 × 10–2 at 500 K (ii) CaCO3 (s) CaO(s) + CO2(g); Kp= 167 at 1073 K

Exercises, continued

Exercises 6.6 to 6.11

- 6.6 For the following equilibrium, Kc= 6.3 × 1014 at 1000 K. NO (g) + O3 (g) NO2 (g) + O2 (g). Both the forward and reverse reactions in the equilibrium are elementary bimolecular reactions. What is Kc, for the reverse reaction?
- 6.7 Explain why pure liquids and solids can be ignored while writing the equilibrium constant expression?
- 6.8 Reaction between N2 and O2 takes place as follows: 2N2 (g) + O2 (g) 2N2O (g). If a mixture of 0.482 mol N2 and 0.933 mol of O2 is placed in a 10 L reaction vessel and allowed to form N2O at a temperature for which Kc= 2.0 × 10–37, determine the composition of equilibrium mixture.
- 6.9 Nitric oxide reacts with Br2 and gives nitrosyl bromide as per reaction given below: 2NO (g) + Br2 (g) 2NOBr (g). When 0.087 mol of NO and 0.0437 mol of Br2 are mixed in a closed container at constant temperature, 0.0518 mol of NOBr is obtained at equilibrium. Calculate equilibrium amount of NO and Br2.
- 6.10 At 450K, Kp= 2.0 × 1010/bar for the given reaction at equilibrium. 2SO2(g) + O2(g) 2SO3 (g). What is Kc at this temperature ?
- 6.11 A sample of HI(g) is placed in flask at a pressure of 0.2 atm. At equilibrium the partial pressure of HI(g) is 0.04 atm. What is Kp for the given equilibrium ? 2HI (g) H2 (g) + I2 (g)

Exercises 6.12 to 6.15

- 6.12 A mixture of 1.57 mol of N2, 1.92 mol of H2 and 8.13 mol of NH3 is introduced into a 20 L reaction vessel at 500 K. At this temperature, the equilibrium constant, Kc for the reaction N2 (g) + 3H2 (g) 2NH3 (g) is 1.7 × 102. Is the reaction mixture at equilibrium? If not, what is the direction of the net reaction?
- 6.13 The equilibrium constant expression for a gas reaction is given. Write the balanced chemical equation corresponding to this expression.
- 6.14 One mole of H2O and one mole of CO are taken in 10 L vessel and heated to 725 K. At equilibrium 40% of water (by mass) reacts with CO according to the equation, H2O (g) + CO (g) H2 (g) + CO2 (g). Calculate the equilibrium constant for the reaction.
- 6.15 At 700 K, equilibrium constant for the reaction: H2 (g) + I2 (g) 2HI (g) is 54.8. If 0.5 mol L–1 of HI(g) is present at equilibrium at 700 K, what are the concentrations of H2(g) and I2(g) assuming that we initially started with HI(g) and allowed it to reach equilibrium at 700K?

Exercises, continued

Exercises 6.16 to 6.19

- 6.16 What is the equilibrium concentration of each of the substances in the equilibrium when the initial concentration of ICl was 0.78 M ? 2ICl (g) I2 (g) + Cl2 (g); Kc = 0.14
- 6.17 Kp = 0.04 atm at 899 K for the equilibrium shown below. What is the equilibrium concentration of C2H6 when it is placed in a flask at 4.0 atm pressure and allowed to come to equilibrium? C2H6 (g) C2H4 (g) + H2 (g)
- 6.18 Ethyl acetate is formed by the reaction between ethanol and acetic acid. (i) Write the concentration ratio (reaction quotient), Qc, for this reaction, noting that water is not in excess and is not a solvent in this reaction. (ii) At 293 K, if one starts with 1.00 mol of acetic acid and 0.18 mol of ethanol, there is 0.171 mol of ethyl acetate in the final equilibrium mixture. Calculate the equilibrium constant. (iii) Starting with 0.5 mol of ethanol and 1.0 mol of acetic acid and maintaining it at 293 K, 0.214 mol of ethyl acetate is found after sometime. Has equilibrium been reached?
- 6.19 A sample of pure PCl5 was introduced into an evacuated vessel at 473 K. After equilibrium was attained, concentration of PCl5 was found to be 0.5 × 10–1 mol L–1. If value of Kc is 8.3 × 10–3, what are the concentrations of PCl3 and Cl2 at equilibrium?

Exercises 6.20 to 6.23

- 6.20 One of the reactions that takes place in producing steel from iron ore is the reduction of iron(II) oxide by carbon monoxide to give iron metal and CO2. FeO (s) + CO (g) Fe (s) + CO2 (g); Kp = 0.265 atm at 1050K. What are the equilibrium partial pressures of CO and CO2 at 1050 K if the initial partial pressures are pCO = 1.4 atm and pCO2 = 0.80 atm?
- 6.21 Equilibrium constant, Kc for the reaction N2 (g) + 3H2 (g) 2NH3 (g) at 500 K is 0.061. At a particular time, the analysis shows that composition of the reaction mixture is 3.0 mol L–1 N2, 2.0 mol L–1 H2 and 0.5 mol L–1 NH3. Is the reaction at equilibrium? If not in which direction does the reaction tend to proceed to reach equilibrium?
- 6.22 Bromine monochloride, BrCl decomposes into bromine and chlorine and reaches the equilibrium: 2BrCl (g) Br2 (g) + Cl2 (g) for which Kc= 32 at 500 K. If initially pure BrCl is present at a concentration of 3.3 × 10–3 mol L–1, what is its molar concentration in the mixture at equilibrium?
- 6.23 At 1127 K and 1 atm pressure, a gaseous mixture of CO and CO2 in equilibrium with soild carbon has 90.55% CO by mass. C (s) + CO2 (g) 2CO (g). Calculate Kc for this reaction at the above temperature.

Exercises, continued

Exercises 6.24 to 6.27

- 6.24 Calculate a) ΔG and b) the equilibrium constant for the formation of NO2 from NO and O2 at 298K. NO (g) + ½ O2 (g) NO2 (g) where ΔfG (NO2) = 52.0 kJ/mol, ΔfG (NO) = 87.0 kJ/mol, ΔfG (O2) = 0 kJ/mol
- 6.25 Does the number of moles of reaction products increase, decrease or remain same when each of the following equilibria is subjected to a decrease in pressure by increasing the volume? (a) PCl5 (g) PCl3 (g) + Cl2 (g) (b) CaO (s) + CO2 (g) CaCO3 (s) (c) 3Fe (s) + 4H2O (g) Fe3O4 (s) + 4H2 (g)
- 6.26 Which of the following reactions will get affected by increasing the pressure? Also, mention whether change will cause the reaction to go into forward or backward direction. (i) COCl2 (g) CO (g) + Cl2 (g) (ii) CH4 (g) + 2S2 (g) CS2 (g) + 2H2S (g) (iii) CO2 (g) + C (s) 2CO (g) (iv) 2H2 (g) + CO (g) CH3OH (g) (v) CaCO3 (s) CaO (s) + CO2 (g) (vi) 4 NH3 (g) + 5O2 (g) 4NO (g) + 6H2O(g)
- 6.27 The equilibrium constant for the following reaction is 1.6 × 105 at 1024K. H2(g) + Br2(g) 2HBr(g). Find the equilibrium pressure of all gases if 10.0 bar of HBr is introduced into a sealed container at 1024K.

Exercises 6.28 to 6.30

- 6.28 Dihydrogen gas is obtained from natural gas by partial oxidation with steam as per the following endothermic reaction: CH4 (g) + H2O (g) CO (g) + 3H2 (g). (a) Write an expression for Kp for the above reaction. (b) How will the values of Kp and composition of equilibrium mixture be affected by (i) increasing the pressure (ii) increasing the temperature (iii) using a catalyst?
- 6.29 Describe the effect of a) addition of H2 b) addition of CH3OH c) removal of CO d) removal of CH3OH on the equilibrium of the reaction: 2H2(g) + CO (g) CH3OH (g)
- 6.30 At 473 K, equilibrium constant Kc for decomposition of phosphorus pentachloride, PCl5 is 8.3 × 10-3. If decomposition is depicted as PCl5 (g) PCl3 (g) + Cl2 (g), ΔrH = 124.0 kJ mol–1, a) write an expression for Kc for the reaction b) what is the value of Kc for the reverse reaction at the same temperature c) what would be the effect on Kc if (i) more PCl5 is added (ii) pressure is increased (iii) the temperature is increased?

Exercises, continued

Exercises 6.31 to 6.36

- 6.31 Dihydrogen gas used in Haber's process is produced by reacting methane from natural gas with high temperature steam. The first stage involves the formation of CO and H2, and in the second stage the CO formed is reacted with more steam in the water gas shift reaction, CO (g) + H2O (g) CO2 (g) + H2 (g). If a reaction vessel at 400°C is charged with an equimolar mixture of CO and steam such that pCO = pH2O = 4.0 bar, what will be the partial pressure of H2 at equilibrium? Kp= 10.1 at 400°C
- 6.32 Predict which of the following reactions will have appreciable concentration of reactants and products: a) Cl2 (g) 2Cl (g) Kc = 5 × 10–39 b) Cl2 (g) + 2NO (g) 2NOCl (g) Kc = 3.7 × 108 c) Cl2 (g) + 2NO2 (g) 2NO2Cl (g) Kc = 1.8
- 6.33 The value of Kc for the reaction 3O2 (g) 2O3 (g) is 2.0 × 10–50 at 25°C. If the equilibrium concentration of O2 in air at 25°C is 1.6 × 10–2, what is the concentration of O3?
- 6.34 The reaction, CO(g) + 3H2(g) CH4(g) + H2O(g) is at equilibrium at 1300 K in a 1L flask. It also contains 0.30 mol of CO, 0.10 mol of H2 and 0.02 mol of H2O and an unknown amount of CH4 in the flask. Determine the concentration of CH4 in the mixture. The equilibrium constant, Kc for the reaction at the given temperature is 3.90.
- 6.35 What is meant by the conjugate acid-base pair? Find the conjugate acid or base for the following species: HNO2, CN–, HClO4, F–, OH–, carbonate ion, and S2–
- 6.36 Which of the following are Lewis acids? H2O, BF3, H+, and NH4+

Exercises 6.37 to 6.43

- 6.37 What will be the conjugate bases for the Brönsted acids: HF, H2SO4 and HCO3–?
- 6.38 Write the conjugate acids for the following Brönsted bases: NH2–, NH3 and HCOO–.
- 6.39 The species H2O, HCO3–, HSO4– and NH3 can act both as Brönsted acids and bases. For each case give the corresponding conjugate acid and base.
- 6.40 Classify the following species into Lewis acids and Lewis bases and show how these act as Lewis acid or base: (a) OH– (b) F– (c) H+ (d) BCl3.
- 6.41 The concentration of hydrogen ion in a sample of soft drink is 3.8 × 10–3 M. What is its pH?
- 6.42 The pH of a sample of vinegar is 3.76. Calculate the concentration of hydrogen ion in it.
- 6.43 The ionization constant of HF, HCOOH and HCN at 298K are 6.8 × 10–4, 1.8 × 10–4 and 4.8 × 10–9 respectively. Calculate the ionization constants of the corresponding conjugate base.

Exercises, continued

Exercises 6.44 to 6.51

- 6.44 The ionization constant of phenol is 1.0 × 10–10. What is the concentration of phenolate ion in 0.05 M solution of phenol? What will be its degree of ionization if the solution is also 0.01M in sodium phenolate?
- 6.45 The first ionization constant of H2S is 9.1 × 10–8. Calculate the concentration of HS– ion in its 0.1M solution. How will this concentration be affected if the solution is 0.1M in HCl also? If the second dissociation constant of H2S is 1.2 × 10–13, calculate the concentration of S2– under both conditions.
- 6.46 The ionization constant of acetic acid is 1.74 × 10–5. Calculate the degree of dissociation of acetic acid in its 0.05 M solution. Calculate the concentration of acetate ion in the solution and its pH.
- 6.47 It has been found that the pH of a 0.01M solution of an organic acid is 4.15. Calculate the concentration of the anion, the ionization constant of the acid and its pKa.
- 6.48 Assuming complete dissociation, calculate the pH of the following solutions: (a) 0.003 M HCl (b) 0.005 M NaOH (c) 0.002 M HBr (d) 0.002 M KOH
- 6.49 Calculate the pH of the following solutions: a) 2 g of TlOH dissolved in water to give 2 litre of solution. b) 0.3 g of Ca(OH)2 dissolved in water to give 500 mL of solution. c) 0.3 g of NaOH dissolved in water to give 200 mL of solution. d) 1mL of 13.6 M HCl is diluted with water to give 1 litre of solution.
- 6.50 The degree of ionization of a 0.1M bromoacetic acid solution is 0.132. Calculate the pH of the solution and the pKa of bromoacetic acid.
- 6.51 The pH of 0.005M codeine (C18H21NO3) solution is 9.95. Calculate its ionization constant and pKb.

Exercises 6.52 to 6.58

- 6.52 What is the pH of 0.001M aniline solution? The ionization constant of aniline can be taken from the table of weak bases. Calculate the degree of ionization of aniline in the solution. Also calculate the ionization constant of the conjugate acid of aniline.
- 6.53 Calculate the degree of ionization of 0.05M acetic acid if its pKa value is 4.74. How is the degree of dissociation affected when its solution also contains (a) 0.01M (b) 0.1M in HCl ?
- 6.54 The ionization constant of dimethylamine is 5.4 × 10–4. Calculate its degree of ionization in its 0.02M solution. What percentage of dimethylamine is ionized if the solution is also 0.1M in NaOH?
- 6.55 Calculate the hydrogen ion concentration in the following biological fluids whose pH are given below: (a) Human muscle-fluid, 6.83 (b) Human stomach fluid, 1.2 (c) Human blood, 7.38 (d) Human saliva, 6.4.
- 6.56 The pH of milk, black coffee, tomato juice, lemon juice and egg white are 6.8, 5.0, 4.2, 2.2 and 7.8 respectively. Calculate corresponding hydrogen ion concentration in each.
- 6.57 If 0.561 g of KOH is dissolved in water to give 200 mL of solution at 298 K. Calculate the concentrations of potassium, hydrogen and hydroxyl ions. What is its pH?
- 6.58 The solubility of Sr(OH)2 at 298 K is 19.23 g/L of solution. Calculate the concentrations of strontium and hydroxyl ions and the pH of the solution.

Exercises, concluded

Exercises 6.59 to 6.66

- 6.59 The ionization constant of propanoic acid is 1.32 × 10–5. Calculate the degree of ionization of the acid in its 0.05M solution and also its pH. What will be its degree of ionization if the solution is 0.01M in HCl also?
- 6.60 The pH of 0.1M solution of cyanic acid (HCNO) is 2.34. Calculate the ionization constant of the acid and its degree of ionization in the solution.
- 6.61 The ionization constant of nitrous acid is 4.5 × 10–4. Calculate the pH of 0.04 M sodium nitrite solution and also its degree of hydrolysis.
- 6.62 A 0.02M solution of pyridinium hydrochloride has pH = 3.44. Calculate the ionization constant of pyridine.
- 6.63 Predict if the solutions of the following salts are neutral, acidic or basic: NaCl, KBr, NaCN, NH4NO3, NaNO2 and KF
- 6.64 The ionization constant of chloroacetic acid is 1.35 × 10–3. What will be the pH of 0.1M acid and its 0.1M sodium salt solution?
- 6.65 Ionic product of water at 310 K is 2.7 × 10–14. What is the pH of neutral water at this temperature?
- 6.66 Calculate the pH of the resultant mixtures: a) 10 mL of 0.2M Ca(OH)2 + 25 mL of 0.1M HCl b) 10 mL of 0.01M H2SO4 + 10 mL of 0.01M Ca(OH)2 c) 10 mL of 0.1M H2SO4 + 10 mL of 0.1M KOH

Exercises 6.67 to 6.73

- 6.67 Determine the solubilities of silver chromate, barium chromate, ferric hydroxide, lead chloride and mercurous iodide at 298K from their solubility product constants given in the table. Determine also the molarities of individual ions.
- 6.68 The solubility product constant of Ag2CrO4 and AgBr are 1.1 × 10–12 and 5.0 × 10–13 respectively. Calculate the ratio of the molarities of their saturated solutions.
- 6.69 Equal volumes of 0.002 M solutions of sodium iodate and cupric chlorate are mixed together. Will it lead to precipitation of copper iodate? (For cupric iodate Ksp = 7.4 × 10–8).
- 6.70 The ionization constant of benzoic acid is 6.46 × 10–5 and Ksp for silver benzoate is 2.5 × 10–13. How many times is silver benzoate more soluble in a buffer of pH 3.19 compared to its solubility in pure water?
- 6.71 What is the maximum concentration of equimolar solutions of ferrous sulphate and sodium sulphide so that when mixed in equal volumes, there is no precipitation of iron sulphide? (For iron sulphide, Ksp = 6.3 × 10–18).
- 6.72 What is the minimum volume of water required to dissolve 1g of calcium sulphate at 298 K? (For calcium sulphate, Ksp is 9.1 × 10–6).
- 6.73 The concentration of sulphide ion in 0.1M HCl solution saturated with hydrogen sulphide is 1.0 × 10–19 M. If 10 mL of this is added to 5 mL of 0.04 M solution of the following: FeSO4, MnCl2, ZnCl2 and CdCl2, in which of these solutions will precipitation take place?